

/ **Bill of Materials (BOM) Electronics (Elektronika)**

- 1x MKS DLC32 v2.1 Motherboard
- 1x Mean Well 24V 200W Power Supply (e.g., LRS-200-24)
- 3x TMC2209 Stepper Motor Drivers
- 3x NEMA 17 Stepper Motors (70mm/48mm body length / high torque) depends how fast and cheap you want to build it
- 1x NEMA 17 Stepper Motor (38mm just be fine)
- 1x AC Power Socket (IEC C14 with switch)
- 3m AC Power Cable (2x1.5mm²)
- 4x Stepper Motor Extension Cables
- 3x Micro Limit Switches / Endstops
- 3x Endstop Cables (2-pin/3-pin)

Motion & Mechanics (Mechanika i Napęd)

- 3x T8x8 Trapezoidal Lead Screws - 500mm length (X/Y axes)
- 1x T8x8 Trapezoidal Lead Screw - 200mm length (Z axis)
- 6x 12mm Linear Smooth Rods/Guides – 500/550mm length (500 be fine but you can use 550)
- 2x 8mm Linear Smooth Rods/Guides - 500mm length
- 2x 8mm Linear Smooth Rods/Guides - 150mm length
- 3x Flexible Shaft Couplers (5mm to 8mm)
- 1x 10/16T GT2 Pulley (5mm bore)
- 1x 20T GT2 Pulley (8mm bore)
- 1x GT2 Closed Loop Timing Belt - 200mm length
- 4x LM12LUU (Long) Linear Bearings
- 4x LM12UU (Short) Linear Bearings
- 8x LM8UU Linear Bearings
- 5x 608ZZ Ball Bearings (8x22x7mm)
- 2x 8mm Shaft Lock Collars / Rings

Frame & Hardware (Konstrukcja)

- 7x 2020 Aluminum Extrusions - 500mm length
- 10x 2020 3-way Corner Brackets (Cube brackets)
- 1x MDF Spoilboard (400x600mm, 18mm thickness)

- Lot x M3 Threaded Heat Set Inserts (OD 4.2mm)
- Lot x Assorted M4/M5 Screws and T-Nuts for 2020 extrusions

Spindle (Wrzeciono)

- 1x Makita 3709 Router (or any other 65mm diameter spindle/trim router)
- I used In Newest Version Parkside Performance PPMF 710 A1 Because it`s cheap and it has a spindle speed control

3D Printing Recommendations (Zalecenia druku)

- It takes about 2kg of filament
- **Material:** PETG, ABS, ASA, or PLA-CF (Do NOT use standard PLA due to stepper motor heat).
- **Walls/Perimeters:** 4 Wall Loops (minimum) for strength.
- **Infill:** 20-40% Gyroid or Cubic. (I used Gyroid)
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You can use basically any 3D printer for this project — just be patient, because it takes a lot of filament and a good amount of time. I printed the parts on a Bambu Lab A1, Bambu Lab X1C, Bambu Lab H2S, and a Creality K1 Max.